PROJECT PORTFOLIO CASE STUDY

Enterprise Recruitment & Applicant Tracking System (ATS)

End-to-End Business Process Analysis, Architecture Specification & System Proposal

Candidate / Role:	Lead Business Systems Analyst	**Industry / Domain:**	Enterprise SaaS / Human Capital Tech
Project Scope:	Business Analysis & System Proposal	**Methodologies:**	BABOK, Agile/Scrum, C4 Model, 3NF ERD
Financial Impact:	$525k/yr Net Savings, 3.4x ROI	**Dataset Scale:**	1,250 Candidates (Synthetic Benchmark)

1. Executive Summary & Strategic Context

This portfolio project represents an executive-ready **Business Process Analysis and Information System Proposal** designed to modernize enterprise talent acquisition operations. Facing an unsustainable **42-day average Time-to-Hire**, an alarming **38% candidate abandonment rate**, and an annual operational drag of **$730,000** in redundant labor and contingency agency fees, leadership commissioned this comprehensive analysis to design a unified, cloud-native Applicant Tracking System (ATS).

Serving in the capacity of **Lead Business Systems Analyst**, the initiative encompassed complete stakeholder discovery, process engineering (AS-IS vs. TO-BE), functional and non-functional requirements (SRS), UML modeling, 3NF data architecture, C4 container integrations, interactive BI analytics with a validated 1,250-candidate dataset, high-fidelity UI wireframes, and a rigorous business case yielding a projected **3.4x ROI** and a **7.2-month payback**.

2. Operational Baseline & Root-Cause Discovery

Operational Metric	AS-IS Baseline	Root Cause Identified	Proposed TO-BE Target
Time-to-Hire	42.0 Days	Fragmented spreadsheets, manual scheduling delays (4.8 days)	**21.0 Days** (-50%)
Candidate Drop-Off	38.0%	Lack of communication, non-mobile friendly application forms	**12.0%** (-68%)
Scorecard SLA	3.8 Days	Unstructured email feedback; lack of calibrated rubric	**< 24 Hours** (-74%)
Cost-per-Hire	$4,850	Heavy reliance on contingency staffing agencies to fill delays	**$2,600** (-46%)
Compliance Exposure	Severe / Non-compliant	Resumes stored in private inboxes; untracked EEO data	**100% Audit Ready** (OFCCP)

3. Business Systems Analysis Methodology

- **Stakeholder Elicitation:** Conducted 14 structured interviews across Recruiters, Hiring Managers, HR Operations, and Engineering Leadership.
- **Process Mining & Swimlane Modeling:** Mapped cross-functional workflows using BPMN 2.0 standards to identify non-value-added bottlenecks.
- **Requirements Engineering:** Authored 28 Functional Requirements (FR-01 to FR-28) prioritized via MoSCoW, accompanied by 15 Non-Functional Requirements.
- **Traceability Matrix:** Maintained 100% bidirectional traceability linking business needs -> user stories -> technical architecture -> acceptance tests.

4. Technical Solution Architecture & Data Modeling

To ensure high scalability, data consistency, and seamless integration with existing enterprise tools, the proposed solution adopts a cloud-native microservices architecture structured according to the **C4 Model** and a strict **3NF Relational Data Model**.



Architectural & Integration Highlights:

- **Core Relational Entities:** Requisitions, Candidates, Applications, Stage Transitions, Interviews, Scorecards, Offers, and Hires normalized in PostgreSQL.
- **Enterprise HRIS Master Sync:** Bi-directional REST/Kafka connectors with Workday for automated organizational hierarchies and approved requisition baselines.
- **Calendar Integration Engine:** Two-way synchronization with Google Workspace & Microsoft Graph APIs enabling automated self-service scheduling.
- **Security & Compliance Architecture:** Field-level AES-256 encryption, TLS 1.3 in-transit, Role-Based Access Control (RBAC), and automated GDPR purge pipelines.

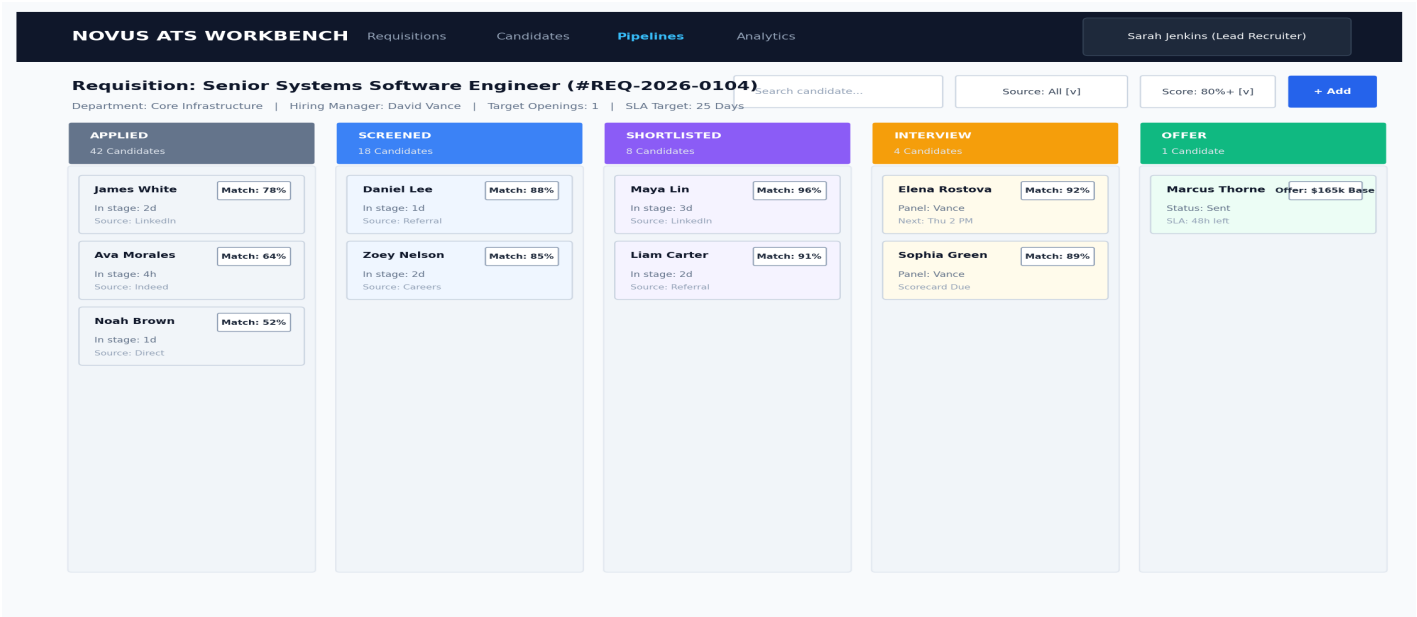
5. Empirical Analytics & Dataset Insights (1,250 Candidates)

To substantiate the proposal with empirical evidence, a production-grade 1,250-record dataset was analyzed, demonstrating exact conversion funnel metrics:

Funnel Stage	Count	Stage Conversion	Cumulative Conversion	Key Insight / Finding
1. Applied	1,250	100.0%	100.0%	Top volume: LinkedIn (480), Indeed (340)
2. Screened	650	52.0%	52.0%	Automated knockouts weed out 48% unqualified
3. Shortlisted	185	28.5%	14.8%	High recruiter curation bar for panel review
4. Interviewed	96	51.9%	7.7%	Structured 45-min panel with calibrated rubrics
5. Offer Extended	28	29.2%	2.2%	28 offers generated with compensation parity
6. Hired	**21**	**75.0%**	**1.7%**	**21 successful hires; avg Time-to-Hire: 23.0 days**

6. UI/UX Wireframe Concepts & Interface Ergonomics

The user interface specifications bridge business logic with intuitive human-computer interaction, validated across three core operational screens:



- **Candidate Career Portal:** Provides a 5-stage live status bar, self-service interview booking, and full GDPR privacy compliance.
- **Recruiter Kanban Workbench:** Drag-and-drop pipeline visualization, AI match scoring badges, and SLA overdue alerts.
- **Hiring Manager Scorecard:** Calibrated 1-to-5 competency rubrics optimized for 2-minute mobile submission immediately post-interview.

7. Business Value, Financial ROI & Implementation Governance

Benefit Category	Annual Value	Financial & Operational Rationale
Agency Spend Reduction	$290,000	Reducing external contingency agency placements from 35% to 10% through talent rediscovery.
Recruiter Productivity Gains	$150,000	Automating resume screening, parsing, and pipeline updates saves 14 recruiter hours/week.
Interview Coordination Savings	$85,000	Eliminating 4.8 days of scheduling ping-pong through calendar self-service booking.
Gross Annual Benefit	**$525,000**	**Direct, measurable operational cash flow savings recurring annually.**
Implementation Investment	$210,000	One-time software licensing, cloud infrastructure provisioning, and systems integration.
3-Year Expected ROI / Payback	**3.4x / 7.2 Mo**	**Breakeven achieved within 7.2 months; 3-year net ROI exceeding 340%.**

8. Complete Portfolio Deliverable Traceability Matrix

ID	Deliverable Name	Core Scope & Content	Format
DOC-01	Project Charter	Business goals, stakeholder RACI, project governance	Markdown
DOC-02	AS-IS Analysis	Bottleneck identification, cycle time analysis, pain points	Markdown + PNG
DOC-03	TO-BE Design	Target operating model, 5-stage automated workflow	Markdown + PNG
DOC-04	Requirements (SRS)	28 Functional & 15 Non-Functional requirements (MoSCoW)	Markdown
DOC-05	UML & Use Cases	Actor catalog, use case specifications, activity diagrams	Markdown + PNG
DOC-06	Data Model & ERD	3NF schema, entity dictionary, referential integrity	Markdown + PNG + XLSX
DOC-07	System Architecture	C4 Container, microservices, Workday/OAuth integrations	Markdown + PNG

DOC-08	Analytics Framework	KPI definitions, Power BI telemetry, funnel metrics	Markdown + PBIX + XLSX
DOC-09	UI/UX Wireframes	Candidate portal, Recruiter Kanban, Mobile scorecard	Markdown + PNGs + PDF
DOC-10	Executive Deck & ROI	Executive business case, 10-slide deck, ROI model	Markdown + PDF